

Lab Title:- SMTP EMAIL TRAFFIC FORENSICS

Student Name:- BADAMASI MUNTAKA MUHAMMAD

Student ID:- 2025FWSD11563

Course Name:- BASIC NETWORKING SKILLS FOR DIGITAL FORENSICS

Instructor Name:- COMMANDANT AMINU IDRIS

Date Of Submission:- 15/09/2026

Executive summary

The lab covers a detailed analysis on smtp email traffic that covers analyzing some of the smtp code/command such as 220, ELO/HELLO and AUTH from the captured smtp traffic it also cover also how to use a python script to decode a encrypted password extracted from the captured smtp traffic.

Lab Objectives

The objectives of the lab is to understand how to conduct an analysis on smtp email traffic.

Tools and Resources Used

- Kali linux:- A penetration testing platform used for security assessments.

Kali linux tools

- tshark:- Dump and analyze network traffic
 - wireshark:- is a GUI network protocol analyzer that lets you interactively browse packet data from a live network or from a previously saved capture file.
 - sha256sum:- Print/compute or check SHA256 (256-bit) checksums message digest.
 - Python 3 base64 module:-use for Decoding Base64 strings offline.
 - Capinfos:- Prints information about capture files.
-

Methology

- ❖ I launched the kali linux terminal.
- ❖ I created a directory and then enter in to the directory
- ❖ I used the w get command to clone/download the smtp email traffic file(smtp.pcap)
- ❖ I made a copy of the smtp.pcap file and the compare it hash value with the original smtp.pcap file using **shasum256** command

- ❖ I used the **capinfos** command to obtain some relevant information about the smtp.pcap file.
- ❖ Using different tshark command option to analyse the smtp.pcap file based on some smtp code/command such as DATA, AUTH,220,TLS/STARTTLS

Screenshot and Evidence

SCREENSHOT SHOWING THE HASH VALUE OF THE CAPURED SMTP TRAFFIC

```
(kali㉿kali)-[~/SBT-DF203-Lab4]
└─$ sudo su
[sudo] password for kali:
(kali㉿kali)-[~/SBT-DF203-Lab4]
└─# sha256sum evidence/smtp.pcap working/smtp_working.pcap | tee reports/smtp_capture_hashes.txt
17ad230db1b6fd5dd18eb311092df1cf6eb162054bdb47697b89bef5a86a47ab evidence/smtp.pcap
17ad230db1b6fd5dd18eb311092df1cf6eb162054bdb47697b89bef5a86a47ab working/smtp_working.pcap
(kali㉿kali)-[~/SBT-DF203-Lab4]
```

SCREENSHOT SHOWING THE CAPINFOS SUMMARY OF THE CAPURED SMTP TRAFFIC

```
(kali㉿kali)-[~/SBT-DF203-Lab4]
└─# capinfos evidence/smtp.pcap | tee reports/smtp_capinfos.txt
File name:          evidence/smtp.pcap
File type:          Wireshark/tcpdump/... - pcap
File encapsulation: Ethernet
File timestamp precision: microseconds (6)
Packet size limit:  file hdr: 65535 bytes
Number of packets:  60
File size:          27 kB
Data size:          26 kB
Capture duration:   9.198384 seconds
Earliest packet time: 2009-10-05 02:06:07.492060
Latest packet time:  2009-10-05 02:06:16.690444
Data byte rate:     2,920 bytes/s
Data bit rate:      23 kbps
Average packet size: 447.77 bytes
Average packet rate: 6 packets/s
SHA256:             17ad230db1b6fd5dd18eb311092df1cf6eb162054bdb47697b89bef5a86a47ab
SHA1:               3def1a1fed849e7f23b66e6925ddff05845a400b
Strict time order:  True
Number of interfaces in file: 1
Interface #0 info:
    Encapsulation = Ethernet (1 - ether)
    Capture length = 65535
    Time precision = microseconds (6)
    Time ticks per second = 1000000
    Number of stat entries = 0
    Number of packets = 60
(kali㉿kali)-[~/SBT-DF203-Lab4]
```

```
(root@kali)~[/home/kali/STB-DF203-Lab4]
# tshark -r working/smtp_working.pcap -q -z conv,tcp | tee reports/tcp_conversations.txt
Running as user "root" and group "root". This could be dangerous.

TCP Conversations
Filter:<No Filter>

|      ←      | |      →      | |      Total      |      Relative      |      Duration      | | | | |
|              | |              | |              | |              | |              | |              |
|              | |              | |              | |              | |              | |              |
|              | |              | |              | |              | |              | |              |
10.10.1.4:1470    ↔ 74.53.140.153:25      25 1,980 bytes      28 22 kB      53 24 kB      0.036986000      7.5
777
```

```
(root@kali) [/home/kali/STB-DF203-Lab4]
# tshark -r working/smtp_working.pcap \
-Y 'smtp.response.code = 220'
Running as user "root" and group "root". This could be dangerous.
6 0.727603 74.53.140.153 → 10.10.1.4 SMTP 235 S: 220-xc90.websitewelcome.com ESMTP Exim 4.69 #1 Mon, 05 Oct 2009 01:05:54 -0500 | We
e use of this system to transport unsolicited, | and/or bulk e-mail.
(root@kali) [/home/kali/STB-DF203-Lab4]
```

```
(root@kali)-[/home/kali/GBT-DF203-Lab4]
# tshark -r working/smtplib_working.pcap \
-Y 'smtp.req.command = "EHLO" || smtp.req.command = "HELO"'

Running as user "root" and group "root". This could be dangerous.
  7   0.732749   10.10.1.4 → 74.53.140.153 SMTP 63 C: EHLO GP

(root@kali)-[/home/kali/GBT-DF203-Lab4]
# tshark -r working/smtplib_working.pcap \
-Y 'smtp.req.command = "AUTH"'

Running as user "root" and group "root". This could be dangerous.
 10   1.076669   10.10.1.4 → 74.53.140.153 SMTP 66 C: AUTH LOGIN

(root@kali)-[/home/kali/GBT-DF203-Lab4]
```

SCREENSHOT SHOWING THE USERNAME AND PASSWORD OF THE CAPTURED SMTP EMAIL TRAFFIC

```
(root@kali)-[/home/kali/ST-DF203-Lab4]
# tshark -r working/smtp_working.pcap -Y 'smtp' -T fields \
  -e frame.number -e frame.time -e ip.src -e tcp.srcport -e ip.dst -e tcp.dstport -e _ws.col.Info \
  | tee reports/smtp_packet_inventory.tsv
Running as user "root" and group "root". This could be dangerous.
6   Oct 5, 2009 02:06:08.219663000 EDT  74.53.140.153  25    10.10.1.4    1470  S: 220-xc90.websitewelcome.com ESMTP Exim 4.69 #1 Mon, 05 Oct 2009
01:05:54 -0500 | We do not authorize the use of this system to transport unsolicited, | and/or bulk e-mail.
7   Oct 5, 2009 02:06:08.224809000 EDT  10.10.1.4    1470  74.53.140.153  25    C: EHLO GP
9   Oct 5, 2009 02:06:08.566183000 EDT  74.53.140.153  25    10.10.1.4    1470  S: 250-xc90.websitewelcome.com Hello GP [122.162.143.157] | SIZE 5
2428800 | PIPELINING | AUTH PLAIN LOGIN | STARTTLS | HELP
10  Oct 5, 2009 02:06:08.568729000 EDT  10.10.1.4    1470  74.53.140.153  25    C: AUTH LOGIN
11  Oct 5, 2009 02:06:08.911081000 EDT  74.53.140.153  25    10.10.1.4    1470  S: 334 VXNlcm5hbWU6
12  Oct 5, 2009 02:06:08.911655000 EDT  10.10.1.4    1470  74.53.140.153  25    C: User: Z3VycGFydGFwQHhhdHJpb3RzLmLu
13  Oct 5, 2009 02:06:09.253544000 EDT  74.53.140.153  25    10.10.1.4    1470  S: 334 UGFzc3dvcmQ6
14  Oct 5, 2009 02:06:09.254118000 EDT  10.10.1.4    1470  74.53.140.153  25    C: Pass: cHVuamFiQDEyMw==
```

SCREENSHOT SHOWING THE BASE64 DECODING RESULT OF THE CAPTURED SMTP TRAFFIC

```
(root@kali)-[/home/kali/ST-DF203-Lab4]
# python3 - <<'PYLAB'
import base64
samples = {
    'username': 'Z3VycGFydGFwQHhhdHJpb3RzLmLu',
    'password': 'cHVuamFiQDEyMw=='
}
for label, value in samples.items():
    try:
        decoded = base64.b64decode(value).decode('utf-8', errors='replace')
        print(f'{label}: {decoded}')
    except Exception as exc:
        print(f'{label}: decode failed: {exc}')
PYLAB
username: gurpartap@patriots.in
password: punjab@123
(root@kali)-[/home/kali/ST-DF203-Lab4]
#
```

SCREENSHOT SHOWING THE MAIL FROM, RCPT TO and DATA OF THE CAPURED SMTP TRAFFIC

```
(root@kali)-[/home/kali/GBT-DF203-Lab4]
# tshark -r working/smtp_working.pcap \
-Y 'smtp.req.command == "MAIL"' \
-T fields \
-e frame.number -e frame.time \
-e ip.src -e ip.dst \
-e smtp.req.command -e smtp.req.parameter

Running as user "root" and group "root". This could be dangerous.
16      Oct  5, 2009 02:06:09.614414000 EDT      10.10.1.4      74.53.140.153      MAIL      FROM: <gurpartap@patriots.in>

(root@kali)-[/home/kali/GBT-DF203-Lab4]
# tshark -r working/smtp_working.pcap \
-Y 'smtp.req.command == "RCPT"' \
-T fields \
-e frame.number -e frame.time \
-e ip.src -e ip.dst \
-e smtp.req.command -e smtp.req.parameter

Running as user "root" and group "root". This could be dangerous.
18      Oct  5, 2009 02:06:09.957250000 EDT      10.10.1.4      74.53.140.153      RCPT      TO: <raj_deol2002in@yahoo.co.in>

(root@kali)-[/home/kali/GBT-DF203-Lab4]
# tshark -r working/smtp_working.pcap \
-Y 'smtp.req.command == "DATA"' \
-T fields \
-e frame.number -e frame.time \
-e ip.src -e ip.dst \
-e smtp.req.command -e smtp.req.parameter

Running as user "root" and group "root". This could be dangerous.
20      Oct  5, 2009 02:06:10.320203000 EDT      10.10.1.4      74.53.140.153      DATA
```

SCREENSHOT SHOWING THE TCP STREAM RECONSTRUCTION OF THE CAPURED SMTP TRAFFIC

```
(root@kali)-[/home/kali/GBT-DF203-Lab4]
# tshark -r working/smtp_working.pcap -q -z follow,tcp,ascii,0 \
tee reports/smtp_stream_STREAM.txt

Running as user "root" and group "root". This could be dangerous.

Follow: tcp,ascii
Filter: tcp.stream eq 0
Node 0: 10.10.1.4:1470
Node 1: 74.53.140.153:25

220-xc90.websitewelcome.com ESMTP Exim 4.69 #1 Mon, 05 Oct 2009 01:05:54 -0500
220-We do not authorize the use of this system to transport unsolicited,
220 and/or bulk e-mail.

9
EHLO GP

137
250-xc90.websitewelcome.com Hello GP [122.162.143.157]
250-SIZE 52428800
250-PIPELINING
250-AUTH PLAIN LOGIN
250-STARTTLS
250 HELP

12
AUTH LOGIN

18
334 VXNlcm5hbWU6

30
```

File Edit View SimpleText Editor Help

```
18
334 VNXlcm5hbWU6
30
Z3VycGFydGFwQHBhdHJpb3RzLmlu

18
334 UGFzc3dvcmQ6

18
chVvamFiQDEyMw==

30
235 Authentication succeeded

36
MAIL FROM: <gurpartap@patriots.in>

8
250 OK

39
RCPT TO: <raj_deol2002in@yahoo.co.in>

14
250 Accepted

6
DATA

56
354 Enter message. ending with "." on a line by itself.

354 Enter message, ending with "." on a line by itself

1460
From: "Gurpartap Singh" <gurpartap@patriots.in>
To: <raj_deol2002in@yahoo.co.in>
Subject: SMTP
Date: Mon, 5 Oct 2009 11:36:07 +0530
Message-ID: <000301ca4581$ef9e57f0$cedb07d0$ain>
MIME-Version: 1.0
Content-Type: multipart/mixed;
    .boundary="=_NextPart_000_0004_01CA4580_095693F0"
X-Mailer: Microsoft Office Outlook 12.0
Thread-Index: AcpFgem9BvjJZEDeR1Kh8i+huYVo0A=
Content-Language: en-us
X-cr-hashedpuzzle: 5eAs AAR2 AdAh BpId C4G1 D1gW FNB1 FBkR Fn+W HFcp HnYJ JO7s Kum6 KytW LFcI LJut;I:cgBHAGoAXwBkAGUAbwBSADIAAAwADIAaQBUEAAeQBhAGgAbwBvBACAAyWbVA
CAAAQBuAA=;Soshal y17;{CAA37F59-1850-45C7-8540-AA27696B5398};ZwB1AHIAcABHIAHIAABHAAQABWAGEAGABYAGKABwB0AHMALgBpAG4A;Mon, 05 Oct 2009 06:06:01 GMT;UwBNAFQAUA=
x-cr-puzzleid: {CAA37F59-1850-45C7-8540-AA27696B5398}

This is a multipart message in MIME format.

-----_NextPart_000_0004_01CA4580_095693F0
Content-Type: multipart/alternative;
    .boundary="=_NextPart_001_0005_01CA4580_095693F0"

-----_NextPart_001_0005_01CA4580_095693F0
Content-Type: text/plain;
    .charset="us-ascii"
Content-Transfer-Encoding: 7bit

Version 4.9.9.1
* Many bug fixes
* Improved editor

Version 4.9.9.0
* Support for latest Mingw compiler system builds
* Bug fixes

Version 4.9.8.9
* New code tooltip display
* Improved Indent/Unindent and Remove Comment
* Improved automatic indent
* Added support for the "interface" keyword
* WebUpdate should now report installation problems from PackMan
* New splash screen and association icons
* Improved installer
* Many bug fixes

Version 4.9.8.7
* Added support for GCC > 3.2
* Debug variables are now reset during next debug session
* Watched Variables not in correct context are now kept and updated when =
    it is needed
* Added new compiler/linker options:-20
    - Strip executable
    - Generate instructions for a specific machine (i386, i486, i586, =
    i686, pentium, pentium-mmx, pentiumpro, pentium2, pentium4,=20
    k6, k6-2, k6-3

1460
, athlon, athlon-tbird, athlon-4, athlon-xp, =
, athlon-mp, winchip-c6, winchip2, k8, c3 and c3-2)
- Enable use of processor specific built-in functions (mmmx, sse, =
```

```

* Saving of custom syntax parameter group
* Possibility of changing compilers and tools filename.
* Many bug fixes

Version 4.9.4.1 (5.0 beta 4.1):
* back to gcc 2.95.3
* Profiling support
* new update/packages checker (vUpdate)
* Lots of bugfixes

--=_NextPart_000_00
04_01CA45B0.095693F0--
-
28
250 OK id=1Mugho-0003Dg-Un
6
QUIT
48
221 xc90.websitewelcome.com closing connection

(root@kali)-[/home/kali/SBT-DF203-Lab4]

```

SCREENSHOT SHOWING THE EMAIL HEADERS AND CLIENT SOFTWARE EXTRACTED FROM THE CAPURED SMTP TRAFFIC

```

(root@kali)-[/home/kali/SBT-DF203-Lab4]
# grep -Ei '^((Date|From|To|Subject|Message-ID|MIME-Version|Content-Type|User-Agent|X-Mailer):)' \
reports/smtp_stream_STREAM.txt | tee reports/message_headers.txt
From: "Gurpartap Singh" <gurpartap@patriots.in>
To: <raj_deol2002in@yahoo.co.in>
Subject: SMTP
Date: Mon, 5 Oct 2009 11:36:07 +0530
Message-ID: <000301ca4581$ef9e57f0$cedb07d0$a@in>
MIME-Version: 1.0
Content-Type: multipart/mixed;
X-Mailer: Microsoft Office Outlook 12.0
Content-Type: multipart/alternative;
Content-Type: text/plain;
Content-Type: text/html;
Content-Type: text/plain;

(root@kali)-[/home/kali/SBT-DF203-Lab4]

```

SCREENSHOT SHOWING THE SOURCE/DESTINATION IP, PORTS & MAC ADDRESSES EXTRACTED FROM THE CAPURED SMTP TRAFFIC

```
(root@kali)-[/home/kali/SBT-DF203-Lab4]
# tshark -r working/smtp_working.pcap -Y 'smtp' -T fields \
  -e frame.number -e frame.time_epoch -e eth.src -e eth.dst -e ip.src -e tcp.srcport \
  -e ip.dst -e tcp.dstport -e tcp.stream \
  | tee reports/smtp_network_metadata.tsv
Running as user "root" and group "root". This could be dangerous.
6      1254722768.21963000      00:1f:33:d9:81:60      00:e0:1c:3c:17:c2      74.53.140.153      25      10.10.1.4      1470      0
7      1254722768.224809000      00:e0:1c:3c:17:c2      00:1f:33:d9:81:60      10.10.1.4      1470      74.53.140.153      25      0
9      1254722768.566183000      00:1f:33:d9:81:60      00:e0:1c:3c:17:c2      74.53.140.153      25      10.10.1.4      1470      0
10     1254722768.568729000      00:e0:1c:3c:17:c2      00:1f:33:d9:81:60      10.10.1.4      1470      74.53.140.153      25      0
11     1254722768.911081000      00:1f:33:d9:81:60      00:e0:1c:3c:17:c2      74.53.140.153      25      10.10.1.4      1470      0
12     1254722768.911655000      00:e0:1c:3c:17:c2      00:1f:33:d9:81:60      10.10.1.4      1470      74.53.140.153      25      0
13     1254722769.253544000      00:1f:33:d9:81:60      00:e0:1c:3c:17:c2      74.53.140.153      25      10.10.1.4      1470      0
14     1254722769.254118000      00:e0:1c:3c:17:c2      00:1f:33:d9:81:60      10.10.1.4      1470      74.53.140.153      25      0
15     1254722769.613798000      00:1f:33:d9:81:60      00:e0:1c:3c:17:c2      74.53.140.153      25      10.10.1.4      1470      0
16     1254722769.614414000      00:e0:1c:3c:17:c2      00:1f:33:d9:81:60      10.10.1.4      1470      74.53.140.153      25      0
17     1254722769.956765000      00:1f:33:d9:81:60      00:e0:1c:3c:17:c2      74.53.140.153      25      10.10.1.4      1470      0
18     1254722769.957250000      00:e0:1c:3c:17:c2      00:1f:33:d9:81:60      10.10.1.4      1470      74.53.140.153      25      0
19     1254722770.319708000      00:1f:33:d9:81:60      00:e0:1c:3c:17:c2      74.53.140.153      25      10.10.1.4      1470      0
20     1254722770.320203000      00:e0:1c:3c:17:c2      00:1f:33:d9:81:60      10.10.1.4      1470      74.53.140.153      25      0
21     1254722770.661679000      00:1f:33:d9:81:60      00:e0:1c:3c:17:c2      74.53.140.153      25      10.10.1.4      1470      0
22     1254722770.692743000      00:e0:1c:3c:17:c2      00:1f:33:d9:81:60      10.10.1.4      1470      74.53.140.153      25      0
23     1254722770.692786000      00:e0:1c:3c:17:c2      00:1f:33:d9:81:60      10.10.1.4      1470      74.53.140.153      25      0
24     1254722770.692804000      00:e0:1c:3c:17:c2      00:1f:33:d9:81:60      10.10.1.4      1470      74.53.140.153      25      0
25     1254722770.692823000      00:e0:1c:3c:17:c2      00:1f:33:d9:81:60      10.10.1.4      1470      74.53.140.153      25      0
26     1254722770.695115000      00:1f:33:d9:81:60      00:e0:1c:3c:17:c2      192.168.1.1,10.10.1.4      1470      10.10.1.4,74.53.140.153      25      0
28     1254722770.695623000      00:1f:33:d9:81:60      00:e0:1c:3c:17:c2      192.168.1.1,10.10.1.4      1470      10.10.1.4,74.53.140.153      25      0
29     1254722770.696248000      00:1f:33:d9:81:60      00:e0:1c:3c:17:c2      192.168.1.1,10.10.1.4      1470      10.10.1.4,74.53.140.153      25      0
30     1254722770.696634000      00:1f:33:d9:81:60      00:e0:1c:3c:17:c2      192.168.1.1,10.10.1.4      1470      10.10.1.4,74.53.140.153      25      0
38     1254722771.494181000      00:e0:1c:3c:17:c2      00:1f:33:d9:81:60      10.10.1.4      1470      74.53.140.153      25      0
39     1254722771.494199000      00:e0:1c:3c:17:c2      00:1f:33:d9:81:60      10.10.1.4      1470      74.53.140.153      25      0
41     1254722771.834628000      00:e0:1c:3c:17:c2      00:1f:33:d9:81:60      10.10.1.4      1470      74.53.140.153      25      0
42     1254722771.834655000      00:e0:1c:3c:17:c2      00:1f:33:d9:81:60      10.10.1.4      1470      74.53.140.153      25      0
44     1254722771.858316000      00:e0:1c:3c:17:c2      00:1f:33:d9:81:60      10.10.1.4      1470      74.53.140.153      25      0
45     1254722771.858334000      00:e0:1c:3c:17:c2      00:1f:33:d9:81:60      10.10.1.4      1470      74.53.140.153      25      0
52     1254722772.248789000      00:1f:33:d9:81:60      00:e0:1c:3c:17:c2      74.53.140.153      25      10.10.1.4      1470      0
```

SCREENSHOT SHOWING THE TLS/STARTTLS ASSESSMENT EXTRACTED FROM THE CAPURED SMTP TRAFFIC

```
(root@kali)-[/home/kali/SBT-DF203-Lab4]
# tshark -r working/smtp_working.pcap \
  -Y 'smtp ||tls'
Running as user "root" and group "root". This could be dangerous.
6      0.727603      74.53.140.153 → 10.10.1.4      SMTP 235 S: 220-xc90.websitewelcome.com ESMTP Exim 4.69 #1 Mon, 05 Oct 2009 01:05:54 -0500 | We do not authorize th
e use of this system to transport unsolicited, | and/or bulk e-mail.
7      0.732749      10.10.1.4 → 74.53.140.153      SMTP 63 C: EHLO GP
9      1.074123      74.53.140.153 → 10.10.1.4      SMTP 191 S: 250-xc90.websitewelcome.com Hello GP [122.162.143.157] | SIZE 52428800 | PIPELINING | AUTH PLAIN LOGIN |
STARTTLS | HELP
10     1.076669      10.10.1.4 → 74.53.140.153      SMTP 66 C: AUTH LOGIN
11     1.419821      74.53.140.153 → 10.10.1.4      SMTP 72 S: 334 VXNlcm5hbWU6
12     1.419595      10.10.1.4 → 74.53.140.153      SMTP 84 C: User: Z3VycGFydGFWQHBhdHJpb3RzLmlu
13     1.761404      74.53.140.153 → 10.10.1.4      SMTP 72 S: 334 UGZzc3Jvcml0eQ==
14     1.762058      10.10.1.4 → 74.53.140.153      SMTP 72 C: Pass: cHVsamF0eQ==
15     2.121738      74.53.140.153 → 10.10.1.4      SMTP 84 S: 235 Authentication succeeded
16     2.122354      10.10.1.4 → 74.53.140.153      SMTP 90 C: MAIL FROM: <gurpartap@patriots.in>
17     2.464705      74.53.140.153 → 10.10.1.4      SMTP 62 S: 250 OK
18     2.465190      10.10.1.4 → 74.53.140.153      SMTP 93 C: RCPT TO: <raj_deol2002in@yahoo.co.in>
19     2.827648      74.53.140.153 → 10.10.1.4      SMTP 68 S: 250 Accepted
20     2.828143      10.10.1.4 → 74.53.140.153      SMTP 60 C: DATA
21     3.169619      74.53.140.153 → 10.10.1.4      SMTP 110 S: 354 Enter message, ending with "." on a line by itself
22     3.200683      10.10.1.4 → 74.53.140.153      SMTP 1514 C: DATA fragment, 1460 bytes
23     3.200726      10.10.1.4 → 74.53.140.153      SMTP 1514 C: DATA fragment, 1460 bytes
24     3.200744      10.10.1.4 → 74.53.140.153      SMTP 1514 C: DATA fragment, 1460 bytes
25     3.200763      10.10.1.4 → 74.53.140.153      SMTP 1514 [TCP Window Full] C: DATA fragment, 1460 bytes
26     3.203055      192.168.1.1 → 10.10.1.4      ICMP 590 Destination unreachable (Fragmentation needed)
28     3.203563      192.168.1.1 → 10.10.1.4      ICMP 590 Destination unreachable (Fragmentation needed)
29     3.204188      192.168.1.1 → 10.10.1.4      ICMP 590 Destination unreachable (Fragmentation needed)
30     3.204574      192.168.1.1 → 10.10.1.4      ICMP 590 Destination unreachable (Fragmentation needed)
38     4.002121      10.10.1.4 → 74.53.140.153      SMTP 1506 C: DATA fragment, 1452 bytes
39     4.002139      10.10.1.4 → 74.53.140.153      SMTP 1506 C: DATA fragment, 1452 bytes
41     4.342568      10.10.1.4 → 74.53.140.153      SMTP 1506 C: DATA fragment, 1452 bytes
42     4.342595      10.10.1.4 → 74.53.140.153      SMTP 1506 C: DATA fragment, 1452 bytes
44     4.366256      10.10.1.4 → 74.53.140.153      SMTP 1506 C: DATA fragment, 1452 bytes
45     4.366274      10.10.1.4 → 74.53.140.153      SMTP/IMF 83 FROM: "gurpartap Singh" <gurpartap@patriots.in>, subject: SMTP, (text/plain) (text/html) (text/plain) (
text/plain) |
52     4.756729      74.53.140.153 → 10.10.1.4      SMTP 82 S: 250 OK id=IMugho-0003Dg-Un
54     7.271765      10.10.1.4 → 74.53.140.153      SMTP 60 C: QUIT
56     7.613407      74.53.140.153 → 10.10.1.4      SMTP 102 S: 221 xc90.websitewelcome.com closing connection
```

Conclusion

In conclusion the lab has help me to understand how to analyse an smtp email traffic and also help me in understanding some of the differences between a normal traffic and an smtp email traffic. It has also add more foundation in regard to how to use tshark to analyse network traffic.

Reference

- ICDFA course Material (SBT-DF203(basic networking skills for digital forensics)- by COMMANDANT AMINU IDRIS
 - ICDFA Recorded sessions
-